• SUSPENSION—VIBRATION FELT AT SPEEDS ABOVE 72 KM/H (45 MPH) • VIBRATION—VIBRATION FELT AT SPEEDS ABOVE 72 KM/H (45 MPH)	Article No. 01-1-14

FORD: 1999-2001 CROWN VICTORIA

MERCURY: 1999-2001 GRAND MARQUIS

ISSUE

Some vehicles may exhibit driveline vibrations at speeds in excess of 72 km/h (45 mph) which may be related to driveshaft, axle, wheels, or their components.

ACTION

Identify the causal component for the vibration and balance or replace as necessary to eliminate the vibration. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Refer to the following Diagnostic Flowcharts (Figures 1-5) as an aid in diagnosing vibration conditions.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 306000, 504000, 509000, 597997, 703000, 703200, 703300, 703400

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1999-2001 CROWN VICTORIA/GRAND MARQUIS
DRIVELINE VIBRATION AT SPEEDS HIGHER THAN 72 KM/H (45 MPH)
WITH AXLE RATIOS 2.73, 3.08, 3.27, 3.55

FOR DEALERSHIPS WITHOUT AN ELECTRONIC VIBRATION ANALYZER

Step No.	Action	Procedure or Specification	If YES	If NO
1	Is vibration felt in driver's seat when driving vehicle with customer at complaint speeds?	Refer to 2001 Workshop Manual, Section 100-04: NVH Diagnosis and Testing.	Go to Step 2.	Refer to 2001 Workshop Manual, Section 100-04: NVH Diagnosis and Testing, Item 7, Diagnostic Procedure.
2	Does vibration continue with transmission shifted into Neutral or when overdrive is off?	Refer to 2001 Workshop Manual, Section 100-04: NVH Diagnosis and Testing.	Go to Step 3.	Refer to 2001 Workshop Manual, Section 100-04: NVH Diagnosis and Testing, Item 7, Diagnostic Procedure.
3	Check driveshaft for visible damage, undercoating, or missing weights.	Refer to 2001 Workshop Manual, Section 100-04: NVH Diagnosis and Testing.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.	Go to Step 4.
4	Check driveshaft vector alignment and torque on bolts.	• Refer to 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft. • Ensure alignment of "white" slash on transmission output shaft and "yellow" index mark on driveshaft.	Go to Step 5.	Reindex driveshaft and retorque bolts per 2001 Workshop Manual, Section 205-01, Removal and Installation of Driveshaft, and go to Step1.
5	Measure driveshaft runout at slip yoke. Within specification?	Specification: 0.889mm (0.035″).	Go to Step 7.	Go to Step 6.
6	Is there excessive vertical play in the spines of the slip yoke?	Specification: 0.254mm (0.010″).	Replace slip yoke or transmission extension housing, whichever is worn, and go to Step 1.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
7	Measure driveshaft runout at middle. Within specification?	Specification: 0.889mm (0.035″).	Go to Step 8.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
8	Field balance driveshaft with maximum two (2) hose clamps. Is vibration eliminated?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, Component Test.	Go to Step 1.	Go to Step 9.
9	Is pinion flange runout within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, Component Test. • Specification: 0.254mm (0.010″).	Go to Step 10.	Replace ring/pinion per 2001 Workshop Manual, Section 205-02: Rear Drive Axle Differential, and go to Step 1.

TB-6258-A

Figure 1 - Article 01-1-14

1999-2001 CROWN VICTORIA/GRAND MARQUIS
DRIVELINE VIBRATION AT SPEEDS HIGHER THAN 72 KM/H (45 MPH)
WITH AXLE RATIOS 2.73, 3.08, 3.27, 3.55

FOR DEALERSHIPS WITHOUT AN ELECTRONIC VIBRATION ANALYZER

Step No.	Action	Procedure or Specification	If YES	If NO
10	Measure driveshaft runout at axle end. Within specification?	Specification: 0.889mm (0.035″).	Go to Step 11.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
11	Are rear wheel runouts within specification?	Refer to 2001 Workshop Manual, Section 204-04: Wheels/Tires.	Go to Step 12.	Correct wheel runout per 2001 Workshop Manual, Section 204-04: Wheels/Tires, and go to Step 1.
12	Are rear tires/wheels balanced?	Refer to 2001 Workshop Manual, Section 204-04: Wheels/Tires.	Go to Step 13.	Correct tire/wheel balance and go to Step 1.
13	Is the axle shaft runout within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, General.	Go to Step 14.	Replace axle shaft per 2001 Workshop Manual, Section 205-02: Rear Drive Axle Differential, and go to Step 1.
14	Are U-joint efforts within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, General.	Go to Step 15.	Replace faulty U-joint per 2001 Workshop Manual, Section 205-00: Driveline System, General, and go to Step 1.
15	Obtain Vibration Analyzer from Field Service Engineering or local dealership for decision diagnosis.	Refer to this TSB, Driveline Vibration with EVA.		

TB-6259-A

Figure 2 - Article 01-1-14

1999-2001 CROWN VICTORIA/GRAND MARQUIS
DRIVELINE VIBRATION AT SPEEDS HIGHER THAN 72 KM/H (45 MPH)
WITH AXLE RATIOS 2.73, 3.08, 3.27, 3.55

FOR DEALERSHIPS WITH AN ELECTRONIC VIBRATION ANALYZER (014-00344)
AND VIBRATE SOFTWARE (215-00003)

Step No.	Action	Procedure or Specification	If YES	If NO
0	Set up first accelometer on upper RH seat stud ("A" input). If equipped with additional accelometer, set it below axle pinion on carrier ("B" input).	Vibrate 4.0 road test procedure.		
1	Are the Hz readings the same for input "A" and "B" when driving vehicle with customer at compliant speeds?	Vibrate 4.0 road test procedure.	Go to Step 2.	Determine next possible source of vibration for input "B" and go to Step 2.
2	Based on the results of the road test, is the vibration vehicle speed related?	Vibrate 4.0 road test procedure.	Go to Step 4.	Go to Step 3.
3	Based on the results of the road test, is the vibration engine speed related?	Vibrate 4.0 road test procedure.	Check current TSB for engine/transmission related vibrations.	Go to Step 1.
4	Is the vibration first order propshaft (driveshaft) related?	Vibrate 4.0 software propshaft speed related vibrations section.	Go to Step 5.	Go to Step 13.
5	Check driveshaft for visible damage, undercoating, or missing weights.	Vibrate 4.0 software propshaft speed related vibrations section.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft.	Go to Step 6.
6	Check driveshaft vector alignment and torque on bolts.	• Refer to 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft. • Ensure alignment of "white" slash on transmission output shaft and "yellow" index mark on driveshaft.	Go to Step 7.	Reindex driveshaft and retorque bolts per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
7	Measure driveshaft runout at slip yoke. Within specification?	Specification: 0.889mm (0.035").	Go to Step 9.	Go to Step 8.
8	Is there excessive vertical play in the spines of the slip yoke?	Specification: 0.254mm (0.010").	Replace slip yoke or transmission extension housing, whichever is worn, and go to Step 1.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
9	Measure driveshaft runout at middle. Within specification?	Specification: 0.889mm (0.035").	Go to Step 10.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.

TB-6249-A

Figure 3 - Article 01-1-14

1999-2001 CROWN VICTORIA/GRAND MARQUIS
DRIVELINE VIBRATION AT SPEEDS HIGHER THAN 72 KM/H (45 MPH)
WITH AXLE RATIOS 2.73, 3.08, 3.27, 3.55

FOR DEALERSHIPS WITH AN ELECTRONIC VIBRATION ANALYZER (014-00344)
AND VIBRATE SOFTWARE (215-00003)

Step No.	Action	Procedure or Specification	If YES	If NO
10	Field balance driveshaft with maximum two (2) hose clamps. Is vibration eliminated?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, Component Test.	Go to Step 1.	Go to Step 11.
11	Is pinion flange runout within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, Component Test. • Specification: 0.254mm (0.010″).	Go to Step 12.	Replace ring/pinion per 2001 Workshop Manual, Section 205-02: Rear Drive Axle Differential, and go to Step 1.
12	Measure driveshaft runout at axle end. Within specification?	Specification: 0.889mm (0.035″).	Go to Step 1.	Replace driveshaft per 2001 Workshop Manual, Section 205-01: Removal and Installation of Driveshaft, and go to Step 1.
13	Is the vibration first order tire related?	Vibrate 4.0 software tire speed related vibrations section.	Go to Step 14.	Go to Step 17.
14	Are rear wheel runouts within specification?	Refer to 2001 Workshop Manual, Section 204-04: Wheels/Tires.	Go to Step 15.	Correct wheel runout per 2001 Workshop Manual, Section 204-04: Wheels/Tires, and go to Step 1.
15	Are rear tires/wheels balanced?	Refer to 2001 Workshop Manual, Section 204-04: Wheels/Tires.	Go to Step 16.	Correct tire/wheel balance and go to Step 1.
16	Is the axle shaft runout within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, General.	Go to Step 1.	Replace axle shaft per 2001 Workshop Manual, Section 205-02: Rear Drive Axle Differential, and go to Step 1.
17	Is the vibration second order propshaft (driveshaft) related?	Vibrate 4.0 software propshaft speed related vibrations section.	Go to Step 18.	Go to Step 1.
18	Are U-joint efforts within specification?	Refer to 2001 Workshop Manual, Section 205-00: Driveline System, General.	Go to Step 1.	Replace faulty U-joint per 2001 Workshop Manual, Section 205-00: Driveline System, General, and go to Step 1.

TB-6250-A

Figure 4 - Article 01-1-14

1999-2001 CROWN VICTORIA/GRAND MARQUIS
DRIVELINE VIBRATION DATA SHEET
WITH AXLE RATIOS 2.73, 3.27, 3.55

Dealership Name: _____ Phone No.: _____

Service Manager: _____ Service Technician: _____

City: _____ State: _____ Fax No.: _____

Last 8 Digits of VIN No.: _____ Vehicle Model: _____ Build Date: _____

Customer Name: _____ Kilometers/Miles: _____

Complaint: _____

Drive Results: _____

Sequence of Repairs: _____

_____

Axle Build Date: _____ (look on last line of metal tag on housing)

Driveshaft color code: _____ (Pink/Pink/Yellow/Yellow – PPYY or Pink/White/Red – PWR or Pink/Green/Blue – PGB or Pink/Blue/Orange – PBO)

Objective Data required for root cause analysis:
1. Note if you do not have a Vibration Analyzer (014-00344), please contact your Field Service Engineer (FSE).
2. Place sensor on upper right hand side seat track on driver's side ("A" input) and place sensor on axle housing pinion ("B" input).

Vibration Reading: Input A: Hz: _____ G: _____ Eng. RPM: _____

Input B: Hz: _____ G: _____ Strobe: _____

Driveshaft Runout				
Location	**Original**	**Replaced**	**Specification**	**Manual Section**
Transmission			0.889mm (0.035″)	205-01
Middle			0.889mm (0.035″)	205-01
Axle			0.889mm (0.035″)	205-01

Pinion Runout at Pilot			
Original	**Replaced**	**Specification**	**Manual Section**
		0.254mm (0.010″)	205-01

Please fax this data sheet to:

John Chevalier, Visteon Customer Service Manager, St. Thomas Assembly Plant
Fax: 519-637-5461 – Phone: 519-637-5250

TB-6260-A

Figure 5 - Article 01-1-14